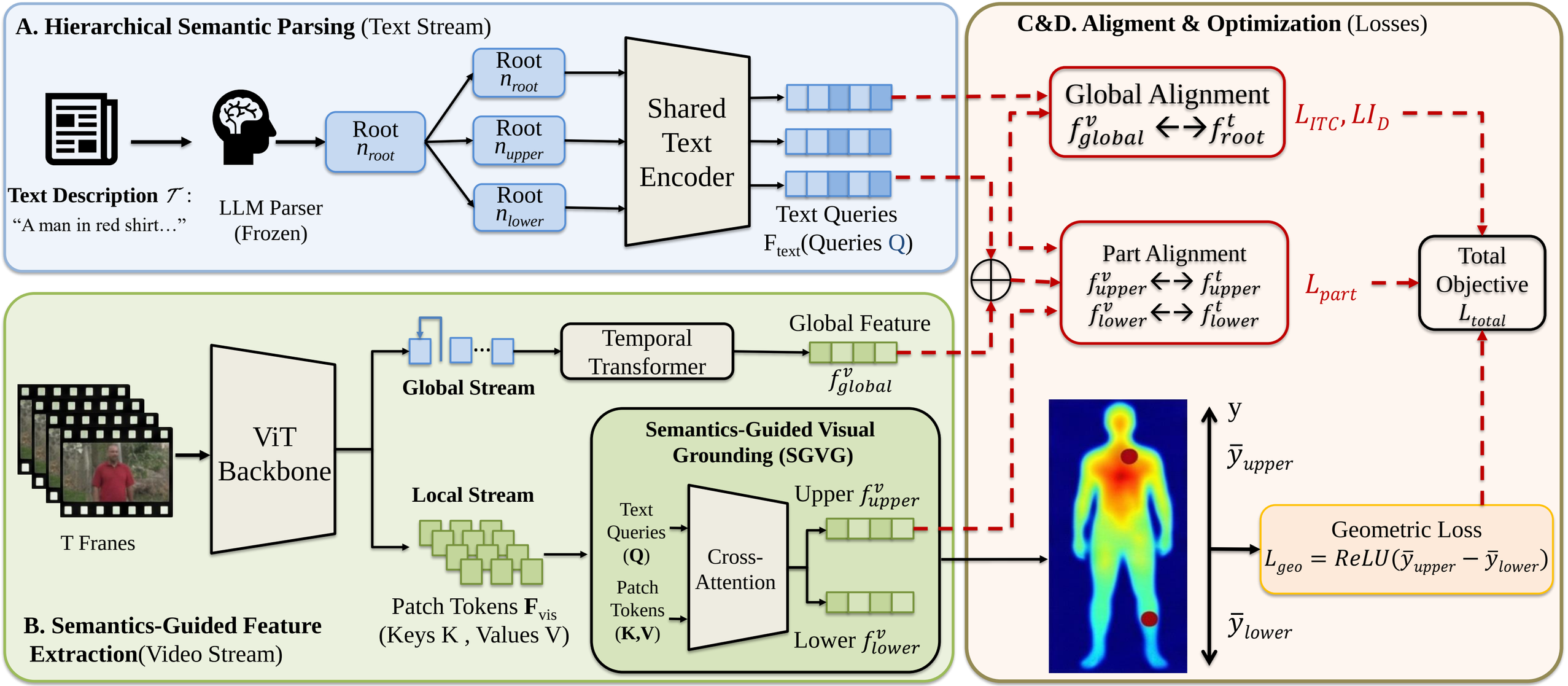
A. Hierarchical Semantic Parsing (Text Stream)
Text Description $\mathcal{T}$ :
"A man in red shirt…"
LLM Parser
(Frozen)
Root $n_{root}$
Root $n_{root}$
Root $n_{upper}$
Root $n_{lower}$
Shared Text Encoder
Text Queries
$F_{text}$(Queries Q)
C&D. Aligment & Optimization (Losses)
Global Alignment
$f^v_{global} \leftrightarrow f^t_{root}$
$L_{ITC}, LI_D$
Part Alignment
$f^v_{upper} \leftrightarrow f^t_{upper}$
$f^v_{lower} \leftrightarrow f^t_{lower}$
$L_{part}$
Total Objective
$L_{total}$
T Franes
ViT Backbone
Global Stream
Temporal Transformer
Global Feature
$f^v_{global}$
Local Stream
Patch Tokens $\mathbf{F}_{vis}$
(Keys K , Values V)
Semantics-Guided Visual Grounding (SGVG)
Text Queries (Q)
Patch Tokens (K,V)
Cross-Attention
Upper $f^v_{upper}$
Lower $f^v_{lower}$
B. Semantics-Guided Feature Extraction(Video Stream)
y
$\bar{y}_{upper}$
$\bar{y}_{lower}$
Geometric Loss
$L_{geo} = ReLU(\bar{y}_{upper} - \bar{y}_{lower})$